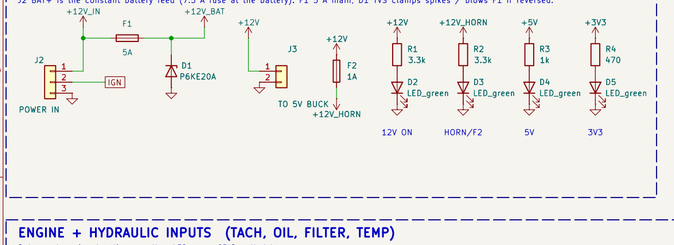
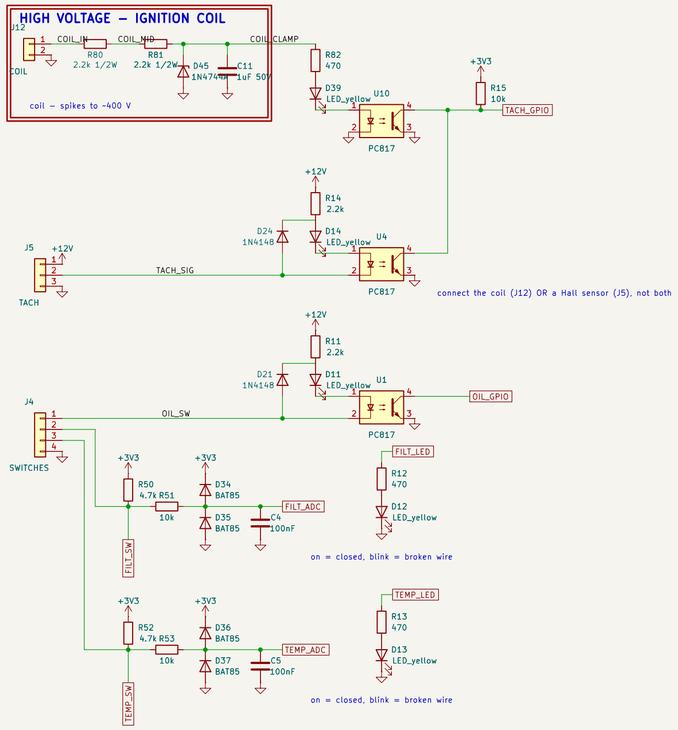
POWER IN / FUSES / PROTECTION / POWER LEDS

J2 BAT+ is the constant battery feed (7.5 A fuse at the battery). F1 5 A main, D1 TVS clamps spikes / blows F1 if reversed.



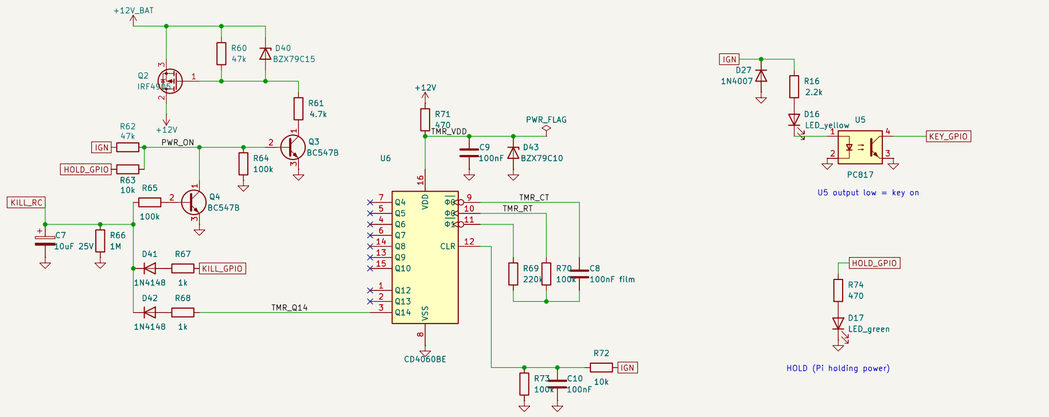
ENGINE + HYDRAULIC INPUTS (TACH, OIL, FILTER, TEMP)

Optocouplers: input active = yellow LED on = GPIO pulled low.
Filter + temp: monitored on the ADC with a 4.7k resistor AT THE SWITCH (1.65 V OK, 0 V closed, 3.3 V wire broken).

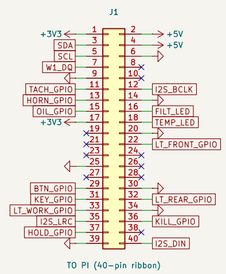


POWER HOLD

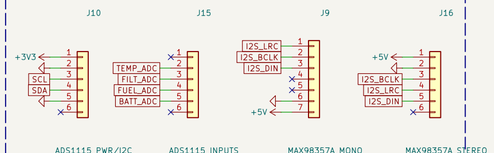
Q2 passes +12V_BAT to +12V while Q3 is on: key on (IGN) OR Pi HOLD high. Q4 cuts it: Pi KILL at the end of shutdown, or the U6 timer ~3 min after key-off. C7 stretches a cut to a few seconds.



TO RASPBERRY PI (40-pin ribbon, pin 1 to pin 1)



PLUG-IN MODULES

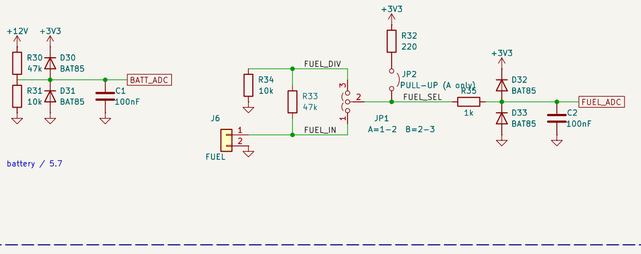


J10 + J15: ADS1115 (Adafruit 1085, two rows 12.7 mm apart)
J10 VIN GND SCL SDA ADDR ALRT; J15 A+ A3 A2 A1 A0 A–
A+ / A– are the module's own analog supply (outputs): left open
J9: MONO amp, Adafruit 3006 (LRC BCLK DIN GAIN SD GND VIN); J16: STEREO amp, Adafruit 6513
(VDD GND BCLK LRC DIN GAIN), Same signals: fit one amp only

Fuel mode: A = sender only: JP1 1–2, fit JP2. B = gauge + dash: JP1 2–3, JP2 off.
Oil + tach: +12V –> 2.2k –> yellow status LED –> PC817 LED –> sensor wire. Switch closes to chassis = LED on = GPIO low.
Filter + temp (monitored): fit a 4.7k resistor across each switch AT THE SWITCH. Wire reads ~1.65 V OK, ~0 V switch closed.
3.3 V wire broken (NO SIGNAL). Their LEDs are driven by the Pi.
F1: 5 A (Pi 4 or Pi 5). D1 (TVS) clamps spikes and blows F1 if power is connected backwards.
Power hold: +12V_IN is the CONSTANT battery feed. Q2 switches it to +12V while the key is on (IGN) or the Pi holds HOLD (GPIO26) high. At the end of shutdown the Pi raises KILL (GPIO16): Q4 cuts power. U6 cuts power ~3 min after key-off if the Pi never does. Key-off draw with everything off: under 0.1 mA.
Lights: sense wires from the LAMP side of each light switch (inline 1 A fuse at the tap). Light on = yellow LED on = GPIO low.
Tach: J12 COIL = ignition coil NEGATIVE terminal (spikes to ~400 V): R80+R81 –> 15 V zener + 1 uF –> PC817 U10.
U10 and U4 (Hall) share TACH_GPIO: connect the coil OR a Hall sensor, not both. 4-cyl coil: pulses_per_rev = 2.
Labels with the same name are connected (flag-shaped = between blocks). GND / +3V3 / +5V at the Pi header and modules are the same supplies as the ground and supply symbols.

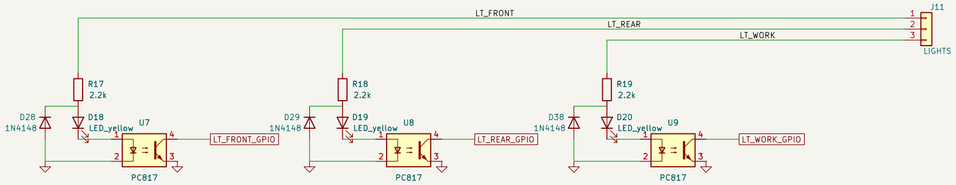
BATTERY / FUEL / BUTTON

ADC inputs clamp to 0..3.3 V. Fuel: A = sender only (JP1 1–2, JP2 on); B = gauge + dash (JP1 2–3, JP2 off).



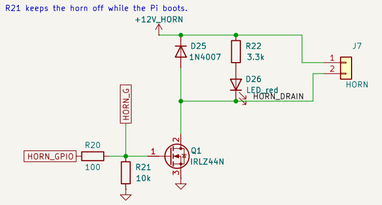
LIGHTS ON (light sensing)

Sense wires from the LAMP side of each light switch (a few mA; the lamps' own current never comes here).



HORN DRIVER

R21 keeps the horn off while the Pi boots.



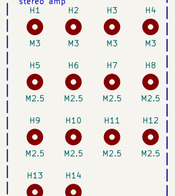
TEST POINTS

Same labels as the board silkscreen.



MOUNTING

H5–H8: ADS1115 standoffs.
H9–H10: mono amp. H11–H14: stereo amp



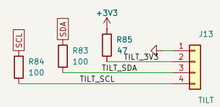
SUPPLY FLAGS (ERC only)

Supplies enter from the harness, the buck and the Pi.



TILT SENSOR CONNECTOR (rev 2.6)

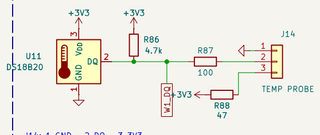
Latching Molex Micro-Fit 3.0, LSM6DSOX board on a short cable (under ~30–50 cm), on the same I2C bus as the ADS1115.



1 GND 2 3V3 3 SDA 4 SCL (cable soldered to the sensor board)

TEMPERATURE (rev 2.7)

1-Wire on GPIO 4. U11 measures the breakout box's air; J14 takes external DS18B20 probes (cab, outside). Several probes can share J14.



J14: 1 GND 2 DQ 3 3V3

Wired schematic: one box per circuit block, the same blocks as the board silkscreen
Harness breakout between the machine wiring and the Raspberry Pi

Sheet: /
File: bobcat-dash-breakout.kicad_sch

Title: Bobcat Digital Dash – Breakout Board

Size: A2	Date: 2026-10-22	Rev: 2.8
KiCad E.D.A. 10.0.7		Id: 1/1